

Introduction to version control with Git - Why we want to track versions and how to go back in time to a working version

 We rewrote this lesson in February and March 2024

If you are looking for the previous version, you can browse the [2023 version of this lesson](#).

This is the introductory lesson to version control using [Git](#).

We start with an **existing repository on the web** to visually explain the basic concepts of version control. We later move to a local repository. Our goal there is not only to be able to apply changes to an existing repository but to also be able to turn own projects into Git repositories and to share them with others.

In the separate [collaborative Git lesson](#), we focus more on the collaboration and the use of remote repositories. We try to stick to simple workflows, just enough for researchers who are not obsessed with Git to be able to work well.

The goals of the module as a whole are that the learner will feel comfortable about committing changes, branching, and merging.

Prerequisites

We offer several options to go through the material: **on the web, in an editor, or in the terminal**. Please see the [installation instructions](#). If you wish to follow in the terminal and are new to the command line, we recorded a [short shell crash course](#).

We recommend to have a [GitHub](#) account. **Why GitHub?** Also [GitLab](#) and [Bitbucket](#) would allow similar workflows and basically everything that we will discuss is transferable. With this material and these exercises we do not implicitly endorse the company [GitHub](#). We have chosen to demonstrate a number of concepts using examples with [GitHub](#) because it is currently the most popular web platform for hosting Git repositories and the chance is high that you will interact with [GitHub](#)-based repositories even if you choose to host your Git repository on another service.

Motivation [1]

Objectives

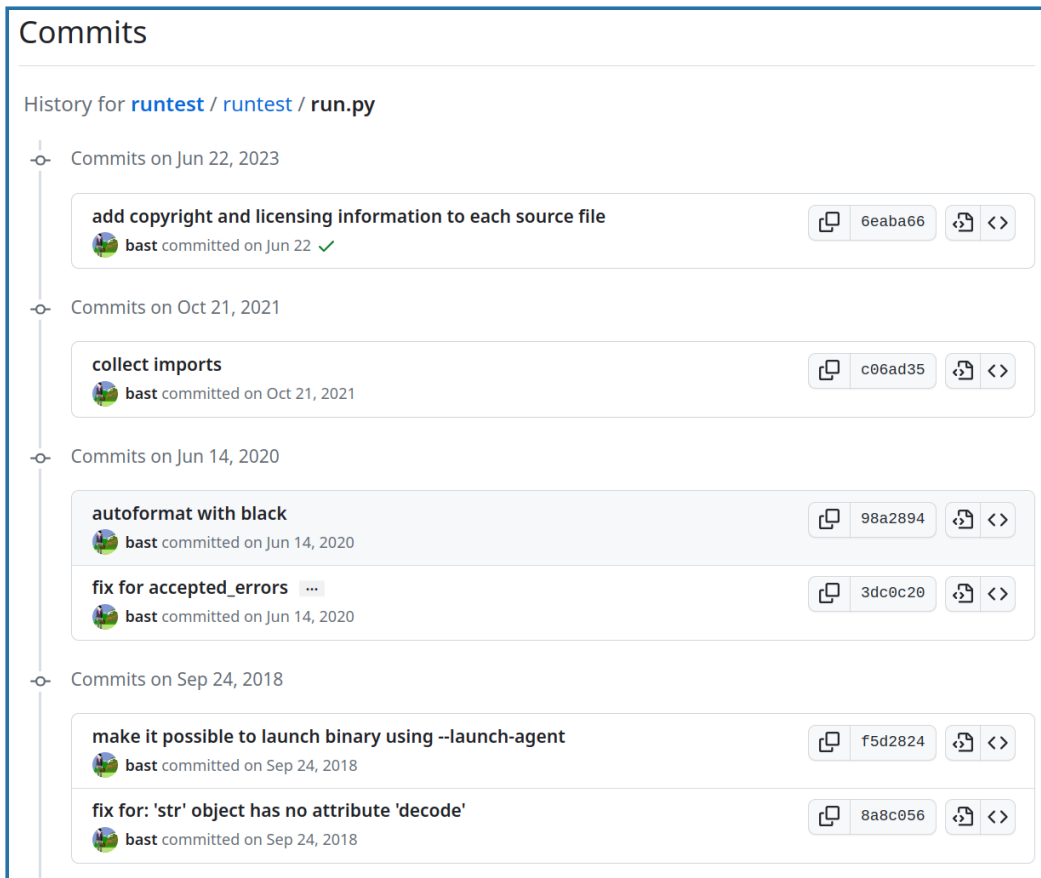
- Make sure nobody leaves the workshop without starting to use some form of version control.
- Discuss the reasons why we advocate distributed version control.

Instructor note

- 15 min discussion/demonstration

Git is all about keeping track of changes

We will learn how to keep track of changes first in the web browser. Below are screenshots of **tracked changes with Git** (from this [example repository](#)):



The screenshot shows the 'Commits' page for the repository 'runtest / runtest / run.py'. It displays a vertical timeline of commits, grouped by date. Each commit entry includes the commit message, the committer's name ('bast'), the commit date, and a hash. There are also icons for cloning, downloading, and viewing the commit details.

Commit Date	Commit Message	Committer	Commit Hash
Jun 22, 2023	add copyright and licensing information to each source file	bast	6eaba66
Oct 21, 2021	collect imports	bast	c06ad35
Jun 14, 2020	autoformat with black	bast	98a2894
Jun 14, 2020	fix for accepted_errors	bast	3dc0c20
Sep 24, 2018	make it possible to launch binary using --launch-agent	bast	f5d2824
Sep 24, 2018	fix for: 'str' object has no attribute 'decode'	bast	8a8c056

Web browser, GitHub view

Later also using the terminal or the editor (the same [example repository](#)):

```
commit 6eaba663debd9790a0f5b8a3ccb9deddd3b5556cb
Author: Radovan Bast <bast@users.noreply.github.com>
Date: Thu Jun 22 17:28:19 2023 +0200

    add copyright and licensing information to each source file

commit c06ad3525060ab249bae44e6d83848b29779eb77
Author: Radovan Bast <bast@users.noreply.github.com>
Date: Thu Oct 21 20:16:33 2021 +0200

    collect imports

commit 98a2894d9f59ae045d117c1bdcc72a6948b3aa49
Author: Radovan Bast <bast@users.noreply.github.com>
Date: Sun Jun 14 16:56:16 2020 +0200

    autoformat with black

commit 3dc0c20f4ebdb519fa11b78aa18392e52e2fbb32
Author: Radovan Bast <bast@users.noreply.github.com>
Date: Sun Jun 14 16:30:57 2020 +0200

    fix for accepted_errors

    previously the code returned 1 before even getting
    to checking accepted/expected errors

commit f5d28243b7daaeff27a06202dcfc3b9b73c9f66c
Author: Radovan Bast <bast@users.noreply.github.com>
Date: Mon Sep 24 19:47:56 2018 +0200

    make it possible to launch binary using --launch-agent

commit 8a8c0563b2f52fb05ad03d806897a740767cf126
Author: Radovan Bast <bast@users.noreply.github.com>
Date: Mon Sep 24 19:38:28 2018 +0200

    fix for: 'str' object has no attribute 'decode'
```

The same as above, but the terminal view

Why do we need to keep track of versions?

Problem: If you have to identify and find your code from 17 days ago, can you?

Version control is an answer to the following questions (do you recognize some of them?):

- “It broke ... hopefully I have a working version somewhere?”
- “Can you please send me the latest version?”
- “Where is the latest version?”
- “Which version are you using?”
- “Which version have the authors used in the paper I am trying to reproduce?”
- “Found a bug! Since when was it there?”
- “I am sure it used to work. When did it change?”
- “My laptop is gone. Is my thesis now gone?”

Features: roll-back, branching, merging, collaboration

Problem: Your code worked two days ago, but is giving an error now. You don't know what you changed.

Problem: You and your colleague want to work on the same code at the same time.

- **Roll-back:** you can always go back to a previous version and compare
- **Branching and merging:**
 - Work on different ideas at the same time
 - Different people can work on the same code/project without interfering
 - You can experiment with an idea and discard it if it turns out to be a bad idea

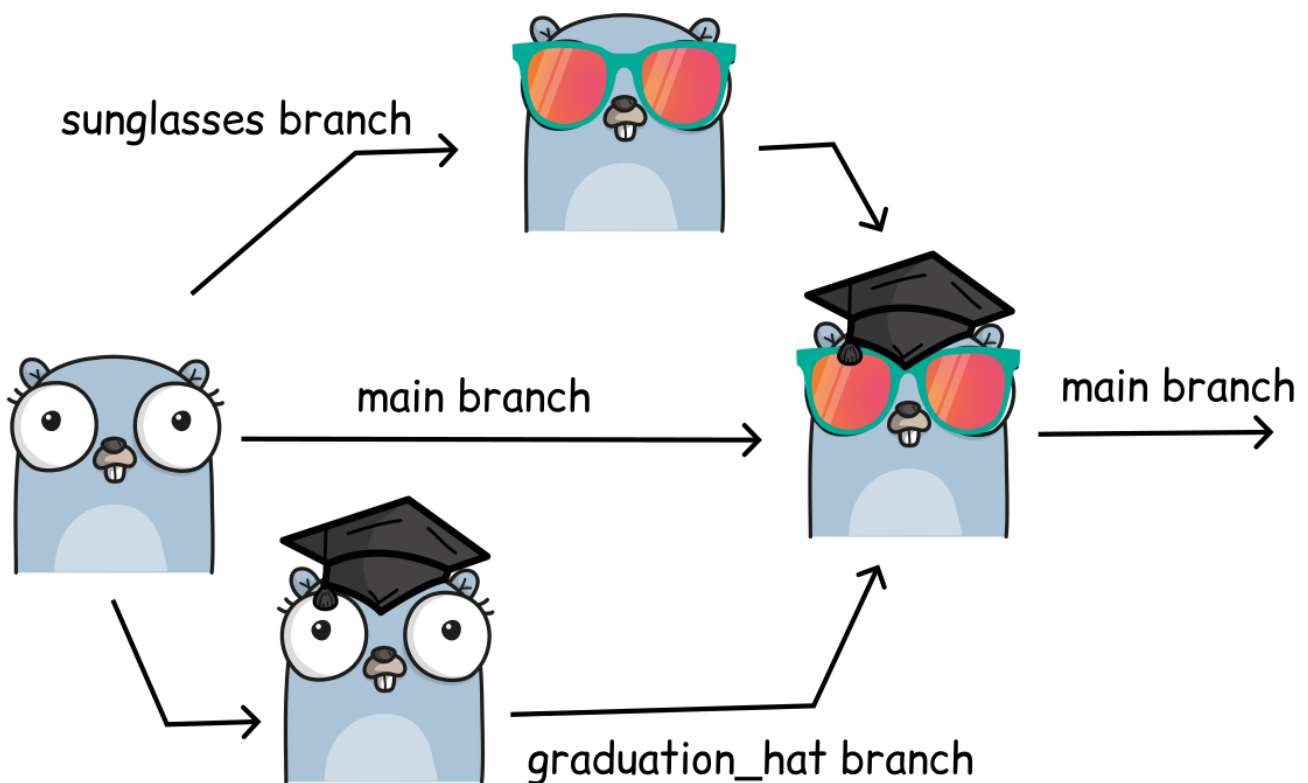


Image created using <https://gopherize.me/> (inspiration).

- **Collaboration:** review, compare, share, discuss
- [Example network graph](#)

Reproducibility

Problem: Someone asks you about your results from 5 years ago. Can you get the same results now?

- How do you indicate which version of your code you have used in your paper?

- When you find a bug, how do you know **when precisely** this bug was introduced (Are published results affected? Do you need to inform collaborators or users of your code?).

With version control we can “annotate” code ([browse this example online](#)):

The screenshot shows the 'Blame' view for the file `boundary.py` in the `networkx/networkx/algorithms` directory. The interface includes a search bar at the top, a commit history bar, and a 'Blame' tab. The code is displayed with line numbers and the commit that introduced each line. The code defines a function `edge_boundary` for finding the boundary of a set of nodes in a graph.

```

1 """Routines to find the boundary of a set of nodes.
2
3 An edge boundary is a set of edges, each of which has exactly one
4 endpoint in a given set of nodes (or, in the case of directed graphs,
5 the set of edges whose source node is in the set).
6
7 A node boundary of a set 'S' of nodes is the set of (out-)neighbors of
8 nodes in 'S' that are outside 'S'.
9
10 """
11 from itertools import chain
12
13 import networkx as nx
14
15 __all__ = ["edge_boundary", "node_boundary"]
16
17
18 @nx._dispatch(edge_attrs="data", preserve_edge_attrs="data")
19 def edge_boundary(G, nbunch1, nbunch2=None, data=False, keys=False, default=None):
20
21     """Returns the edge boundary of `nbunch1`.
22
23     The *edge boundary* of a set *S* with respect to a set *T* is the
24     set of edges (*u*, *v*) such that *u* is in *S* and *v* is in *T*.
25     If *T* is not specified, it is assumed to be the set of all nodes
26     not in *S*.
27
28     Parameters
29     -----
30     G : NetworkX graph
  
```

Example of a git-annotated code with code and history side-by-side.

Talking about code

Problem: You want to show someone a few lines from one of your projects.

Which of these two is more practical?

- “Clone the code, go to the file ‘src/util.rs’, and search for ‘time_iso8601’”. Oh! But make sure you use the version from August 2023.”
- Or I can send you a [permlink](#):

```

37  #[cfg(test)]
38  pub(crate) use set;
39
... 40  // Get current time as an ISO time stamp.
41  pub fn time_iso8601() -> String {
42      let local_time = Local::now();
43      format!("{}", local_time.format("%Y-%m-%dT%H:%M:%S%Z"))
44  }
45
46  // Carve up a line of text into space-separated chunks + the start indices of the chunks.
47  ✓ pub fn chunks(input: &str) -> (Vec<usize>, Vec<&str>) {
48      let mut start_indices: Vec<usize> = Vec::new();

```

Permalink that points to a code portion.

What we typically like to snapshot

- Software (this is how it started but Git/GitHub can track a lot more)
- Scripts
- Documents (plain text files much better suitable than Word documents)
- Manuscripts (Git is great for collaborating/sharing LaTeX or [Quarto](#) manuscripts)
- Configuration files
- Website sources
- Data

Motivation [2]

Difficulties of version control

Despite the benefits, let's be honest, there are some difficulties:

- One more thing to learn (it's probably worth it and will save you more time in the long run; basic career skill).
- Difficult if your collaborators don't want to use it (in the worst case, you can version control **on your side** and email them versions).
- Advanced things can be difficult, but basics are often enough (ask others for help when needed).

Why Git and not another tool?

- **Easy to set up:** no server needed.
- **Very popular:** chances are high you will need to contribute to somebody else's code which is tracked with Git.
- **Distributed:** good backup, no single point of failure, you can track and clean-up changes offline, simplifies collaboration model for open-source projects.
- Important **platforms** such as [GitHub](#), [GitLab](#), and [Bitbucket](#) build on top of Git.

However, any version control is better than no version control and it is OK to prefer a different tool than Git such as [Subversion](#), [Mercurial](#), [Pijul](#), or others.

Configuring Git command line and editor

We have a longer version of this in the [installation instructions](#). But for clarity, we will review the most important parts here.

You don't need to set these if you work only through the GitHub web interface. If you use VS Code or other editors or integrated development environments, the editor might prompt you to set these up. If you use RStudio, you can carry out the configuration in RStudio (see the RStudio tab in the section [Authenticating to GitHub](#) below).

These configuration settings are saved in a file called `.gitconfig` in your home directory. If this file exists, editors like VS Code will use this configuration.

If you want to see your configuration settings, you can use the command (`--show-origin` means it shows the file *where* each setting is defined):

```
$ git config --list --show-origin
```

Name and email address for Git commit metadata

Git commits carry metadata about the author and two things you will always need to define somewhere are:

```
$ git config --global user.name "Your Name"
$ git config --global user.email yourname@example.com
```

For the email address we recommend to use the one you use for your GitHub account. If you prefer to not use it, you can instead use `YOUR_GITHUB_USERNAME@users.noreply.github.com` as the email address (replace `YOUR_GITHUB_USERNAME`). This means that nobody can write to this email address, but GitHub will still be able to connect your contributions with your GitHub account.

Note that these can, in theory, be anything: this is just data, not a registration or identity requirement.

Default branch name

The default branch name in Git has been `master` for a long time, but it is changing to `main` in many places. We recommend to set it to `main` for new repositories that you create locally:

```
$ git config --global init.defaultbranch main
```

Useful alias for the command line

We recommend to define an alias (shortcut) in Git, to be able to nicely visualize branch structure in the terminal without having to remember a long Git command:

```
$ git config --global alias.graph "log --all --graph --decorate --oneline"
```

We have an own section about aliases: [aliases](#).

Default text editor for commit messages

Git sometimes needs to start a text editor for you to enter messages (unless you create commits from inside an editor or on the web). This may have already been set to something (like VS Code), but if not `nano` is usually a safe choice:

```
$ git config --global core.editor nano
```

The [installation instructions text editor page](#) gives ways to set other editors, or do a web search for “git set editor to [editor name]”.

Authenticating to GitHub: SSH or HTTPS or VS Code or RStudio?

How does GitHub know who you are? We discuss here four options:

- **SSH** is the classic method, using [Secure Shell Protocol](#) remote connection keys.
- **HTTPS** works with the **Git Credential Manager**, which is an extra add-on that works easily in Windows and Mac.
- **VS Code** editor can authenticate with GitHub using its own authentication method.
- **RStudio** can handle authentication via HTTPS or SSH. Here, the RStudio instructions cover using **HTTPS**.

Read how to install them from the [installation instructions](#).

Test which one you should use:

Command line: SSH

Command line: HTTPS

VS Code

RStudio

Try this command:


```
$ ssh -T git@github.com
```

If it returns `Hi USERNAME! You've successfully authenticated, ...`, then SSH is configured and the following steps will work with the SSH cloning.

See our [installation instructions](#) to set up SSH access.

From now on, if you know that SSH works, you should always select SSH as the clone URL from GitHub, or translate the URL to start with the right thing yourself:

`git@github.com:` (with the `:`).

Copy and browse an existing project [1]

In this episode, we will look at an **existing repository** to understand how all the pieces work together. Along the way, we will make a copy (a fork) of the repository for us, which will be used for our own changes in the next episode.

- We used to start by directly going and creating a repository from scratch. This was abstract and hard to understand.
- Instead, we'll show you all the cool stuff in a Git repository first, and then start adding files.
- We use an example recipe book we created just for this course.
- By the end of the course, you'll know how to contribute your own recipes to it.

📌 Objectives

- See a real Git repository and understand what is inside of it.
- Understand how version control allows advanced inspection of a repository.
- See how Git allows multiple people to collaborate relatively easily.
- **See the big picture** instead of remembering a bunch of commands.

GitHub, VS Code, Command line, and more

We offer **four different paths** for this exercise:

- **GitHub** (this is the one we will demonstrate on day 1)
- **VS Code** (if you prefer to follow along using an editor; we will return to this on day 2)
- **RStudio** (if you prefer to follow along using an editor; we will return to this on day 2)
- **Command line** (for people comfortable with the command line; you will see more of this on day 2)

In the future we'll add more paths, for example Jupyter ([contributions welcome!](#)).

Creating a copy of the repository by “forking”

A repository is a collection of files in one directory tracked by git. A GitHub repository is GitHub’s copy, which adds things like access control. Each GitHub repository is owned by a user or organization, who controls access.

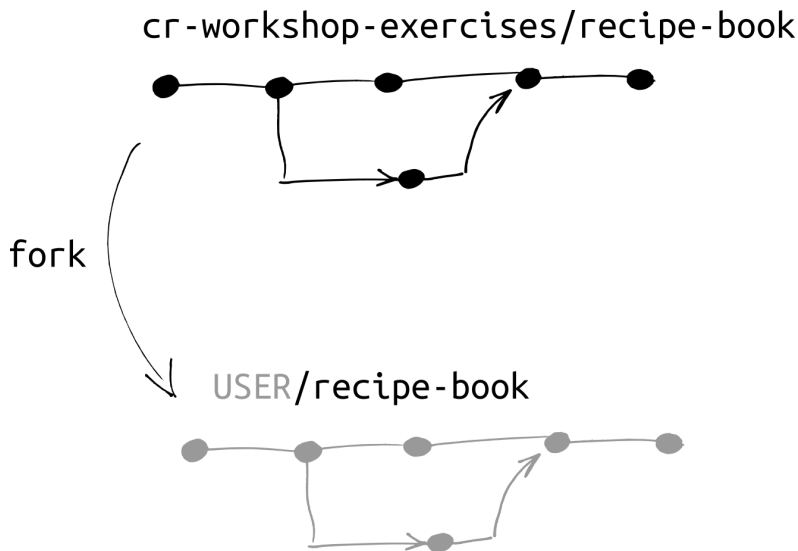


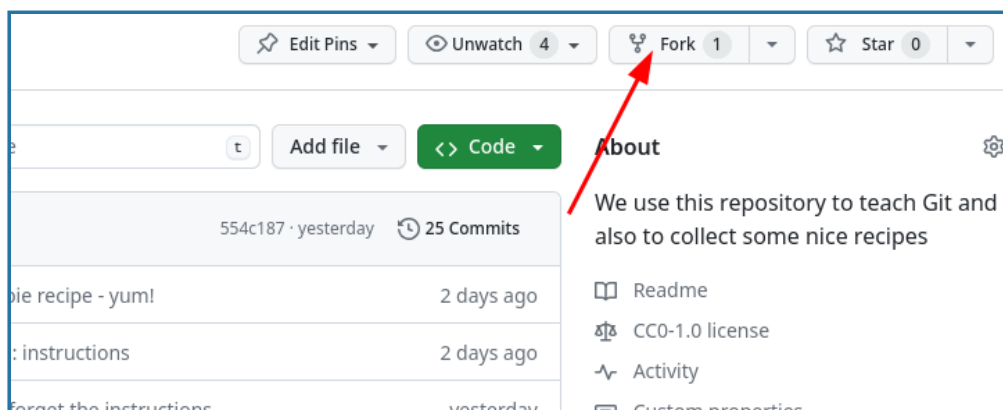
Illustration of forking a repository on GitHub.

First, we need to make **our own copy** of the exercise repository. This will become important later, when we make our own changes.

1. Go to the repository view on GitHub:

- <https://github.com/cr-workshop-exercises/recipe-book>: you can use this one if you don’t want your fork and contributions to be visible on the stream or the recording
- <https://github.com/cr-workshop-exercises/recipe-book-recorded>: we will use this one for the demonstration which is streamed and recorded

2. First, on GitHub, click the button that says “Fork”. It is towards the top-right of the screen:



3. You should shortly be redirected to your copy of the repository **USER/recipe-book**.

At all times you should be aware of if you are looking at *your* repository or the *CodeRefinery upstream* repository.

- Your repository: <https://github.com/USER/recipe-book>

- CodeRefinery upstream repository: <https://github.com/cr-workshop-exercises/recipe-book>

GitHub

VS Code

Command line

RStudio

You only need to open your own view, as described above. The browser URL should look like `https://github.com/USER/recipe-book`, where `USER` is your GitHub username.

Copy and browsing an existing project [2]

Exercise

Work on this by yourself or in your team.

Instructor note

Before starting the exercise session show how to fork the repository to own account (above).



Exercise: Browsing an existing project (25 min)

Browse the recipe-book project (introduced above) and explore commits and branches. Take notes and prepare questions. The hints are for the GitHub path in the browser.

1. Browse the **commit history**: Are commit messages understandable? (Hint: “Commit history”, the timeline symbol, above the file list)
2. Compare the commit history with the **network graph** (“Insights” -> “Network”). Can you find the branches?
3. How can you find out when a recipe was **last modified**?
4. **How many changes** did the Guacamole recipe receive (you find it under “sides”)? Try to click on some of the commits to see what changed. (Hint: “History” in the view of a single file)
5. **Which recipes include the ingredient “salt”**? (Hint: the GitHub search. From the repository view, it should offer the filter “repo:USER/recipe-book” by default. What if you add a search term?)
6. In the Guacamole recipe, find out **who modified each line last and when** (click on file, then click “Blame” button). Find out who added the cilantro and in which commit. (Hint: “Blame” view in the file view)
7. Can you use these recipes yourself? **Are you allowed to share modifications**? (Hint: look for a license file)

8. **Browse issues and pull requests** in the upstream repository (the repository you forked from). Any idea what these might be good for? (Hint: tabs in the repository view)

The solution below goes over most of the answers, and you are encouraged to use it when the hints aren't enough - this is by design.

Solution and walk-through

(1) Basic browsing

The most basic thing to look at is the history of commits.

- This is visible from a button in the repository view. We see every change, when, and who has committed.
- Every change has a unique identifier, such as `554c187`. This can be used to identify both this change, and the whole project's version as of that change.
- Clicking on a change in the view shows more.

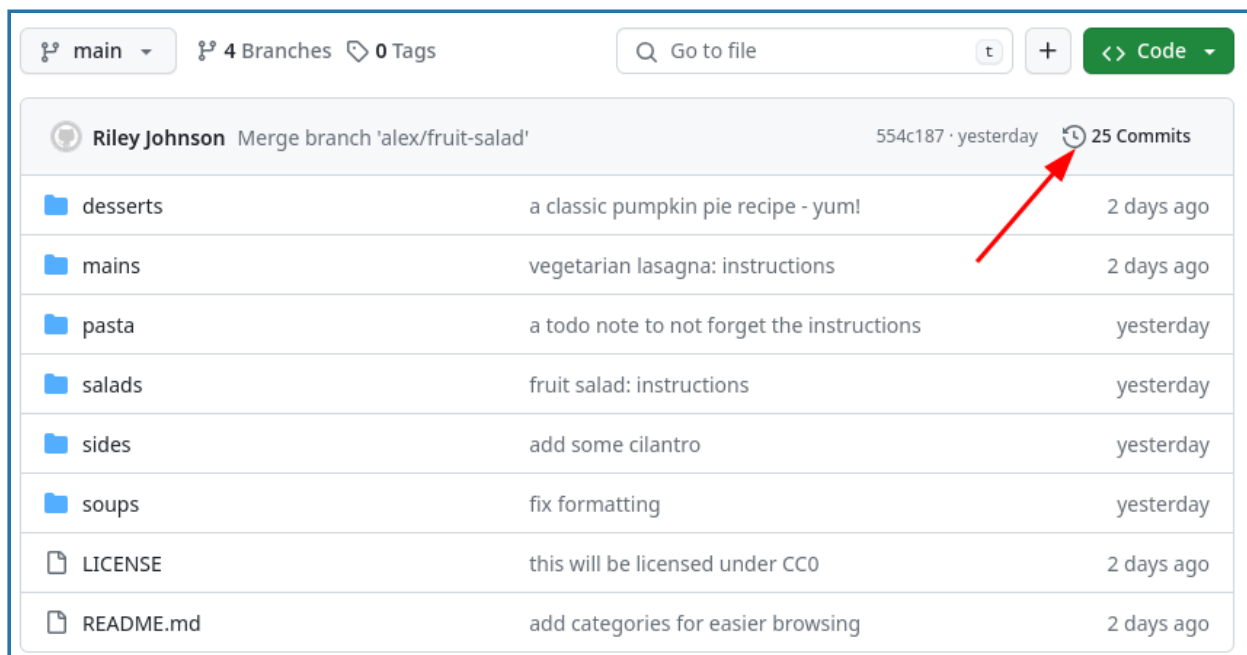
GitHub

VS Code

Command line

RStudio

Click on the timeline symbol in the repository view:



main	4 Branches	0 Tags	Go to file	+	<> Code
Riley Johnson	Merge branch 'alex/fruit-salad'	554c187 · yesterday	🕒	25 Commits	
📁 desserts	a classic pumpkin pie recipe - yum!			2 days ago	
📁 mains	vegetarian lasagna: instructions			2 days ago	
📁 pasta	a todo note to not forget the instructions			yesterday	
📁 salads	fruit salad: instructions			yesterday	
📁 sides	add some cilantro			yesterday	
📁 soups	fix formatting			yesterday	
📄 LICENSE	this will be licensed under CC0			2 days ago	
📄 README.md	add categories for easier browsing			2 days ago	

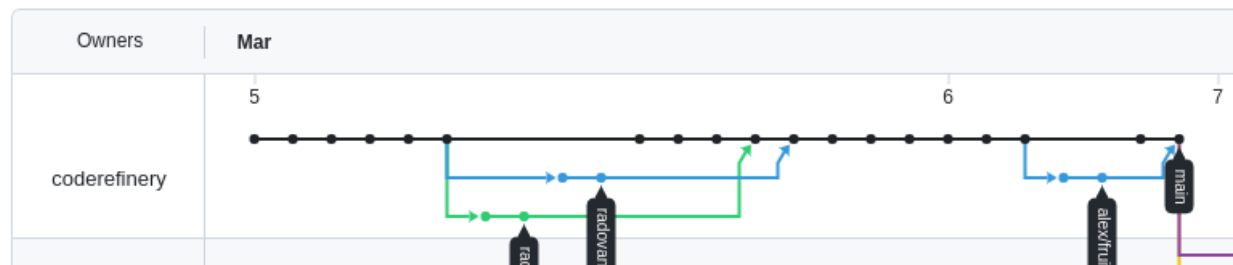
(2) Compare commit history with network graph

The commit history we saw above looks linear: one commit after another. But if we look at the network view, we see some branches and merges. We'll see how to do these later. This is another one of the basic Git views.

In a new browser tab, open the “Insights” tab, and click on “Network”. You can hover over the commit dots to see the person who committed and how they correspond with the commits in the other view:

Network graph

Timeline of the most recent commits to this repository and its network ordered by most recently pushed to.



(3) When was a recipe last modified?

We see the history for the whole repository, but we can also see it for a single file.

Navigate to the file view: Main page → sides directory → guacamole.md. Click the “History” button near the top right:

History for [recipe-book](#) / [sides](#) / [guacamole.md](#) on [main](#)

All users

All time

Commits on Mar 6, 2024

add some cilantro

7051cca

Jamie Garcia committed yesterday

Commits on Mar 5, 2024

reduce amount of salt

a550963

bast committed 2 days ago

don't forget to enjoy

f2d6d58

bast committed 2 days ago

add half an onion

1fde064

bast committed 2 days ago

drafting a guacamole recipe

4c1873e

bast committed 2 days ago

End of commit history for this file

(4) How many changes did the Guacamole recipe receive?

According to the view above, it seems to have five changes (as of 2024-03-07). This could change later on.

(5) Which recipes include the ingredient “salt”

Version control makes it very easy to find all occurrences of a single word. This is useful for things like finding where functions or variables are defined or used.

GitHub

VS Code

Command line

RStudio

We go to the main recipe book view. We click the Search magnifying glass at the very top, type “salt”, and click enter. We see every instance, including the context.

❗ Searching in a forked repository will not work instantaneously!

It usually takes a few minutes before one can search for keywords in a forked repository since it first needs to build the search index the very first time we search. Start it, continue with other steps, then come back to this.

Filter by

- <> Code 5
- Issues 0
- Pull requests 0
- Discussions 0
- Commits 1
- Packages 0
- Wikis 0

Paths

- desserts/
- mains/
- salads/
- sides/
- soups/
- More directories...

Advanced

- Owner
- Symbol
- Exclude archived
- Advanced search

5 files (64 ms) in coderefinery/recipe-book

salads/mixed.md

```

19 - **1 small clove** garlic, minced
20 - **Salt and pepper** to taste
21
26 - In a small bowl, whisk together the olive oil, lemon juice, honey (or
27   maple syrup), minced garlic, salt, and pepper until well combined. Set
28   aside.

```

sides/guacamole.md

```

8 - 1 lime
9 - 1 tsp salt
10 - 1/2 onion
19 - squeeze lime
20 - add salt
21 - and mix well

```

soups/mushroom-soup.md

```

12 - **1 cup** (240ml) heavy cream or coconut milk for a vegan option
13 - **Salt and black pepper** to taste
14 - **2 tablespoons** all-purpose flour (optional, for thickening)
46 6. **Add Cream:**
47 - Stir in the heavy cream or coconut milk and season with salt and pepper to
48   taste. Heat the soup until it is just warmed through; do not boil after

```

Show 1 more match

desserts/pumpkin-pie.md

```

15 - Nutmeg
16 - Can of evaporated milk
17 - 0.4 cup sugar
18 - 0.5 tsp salt
19 - 2 tsp cinnamon

```

(6) Who modified each line last and when?

This is called the “annotate” or “blame” view. The name “blame” is very unfortunate, but it is the standard term for historical reasons for this functionality and it is not meant to blame anyone.

GitHub

VS Code

Command line

RStudio

From a recipe view, change preview to “Blame” towards the top-left. To get the actual commit, click on the commit message.

recipe-book / sides / guacamole.md

Jamie Garcia add some cilantro 7051cca · yesterday History

Preview Code **Blame** 22 lines (16 loc) · 223 Bytes Raw Copy Download Edit

Older Newer Contributors 1

Commit	Author	Message	Line	Code
2 days ago	drafting a guacamole recipe		1	# Guacamole recipe
			2	
			3	
			4	## Ingredients
			5	
			6	- 2 avocados
			7	- 1 chili
			8	- 1 lime
2 days ago	reduce amount of salt		9	- 1 tsp salt
2 days ago	add half an onion		10	- 1/2 onion
yesterday	add some cilantro		11	- 1 Tbsp cilantro
2 days ago	drafting a guacamole recipe		12	
			13	
			14	## Instructions
			15	
			16	- chop avocados
			17	- chop onion
			18	- chop chili
			19	- squeeze lime
			20	- add salt
			21	- and mix well
2 days ago	don't forget to enjoy		22	- enjoy!

(7) Can you use these recipes yourself? Are you allowed to share modifications?

- Look at the file `LICENSE`.
- It says it is “Creative Commons Zero 1.0”, which is equivalent to public domain. You can use them without conditions.
- Note the GitHub view of the file `LICENSE` gives a nice summary of what it means. Try it out:

recipe-book / LICENSE

coderefinery/recipe-book is licensed under the

Creative Commons Zero v1.0 Universal

The Creative Commons CC0 Public Domain Dedication waives copyright interest in a work you've created and dedicates it to the world-wide public domain. Use CC0 to opt out of copyright entirely and ensure your work has the widest reach. As with the Unlicense and typical software licenses, CC0 disclaims warranties. CC0 is very similar to the Unlicense.

This is not legal advice. [Learn more about repository licenses](#)

Permissions	Limitations	Conditions
✓ Commercial use	✗ Liability	
✓ Modification	✗ Trademark use	
✓ Distribution	✗ Patent use	
✓ Private use	✗ Warranty	

(8) Browse issues and pull requests in the upstream repository

This can only be done through the GitHub view. Go to the main repository `cr-workshop-exercises/recipe-book`, (not your fork): <https://github.com/cr-workshop-exercises/recipe-book>. Issues and Pull requests are different for each GitHub copy.

- Click on the “Issues” tab. These are notes that people have added, which allow discussion about the project. Often they are used to communicate problems or ideas.
- Click on the “Pull requests” tab. This allows anyone to *propose changes*, but only the repository owners can accept.

Summary

- Git allowed us to understand this simple project much better than we could, if it was just a few files on our own computer.
- It was easy to share the project with the course.
- By forking the repository, we created our own copy. This is important for the next episode, where we will make changes to our copy.

Recording changes [1]

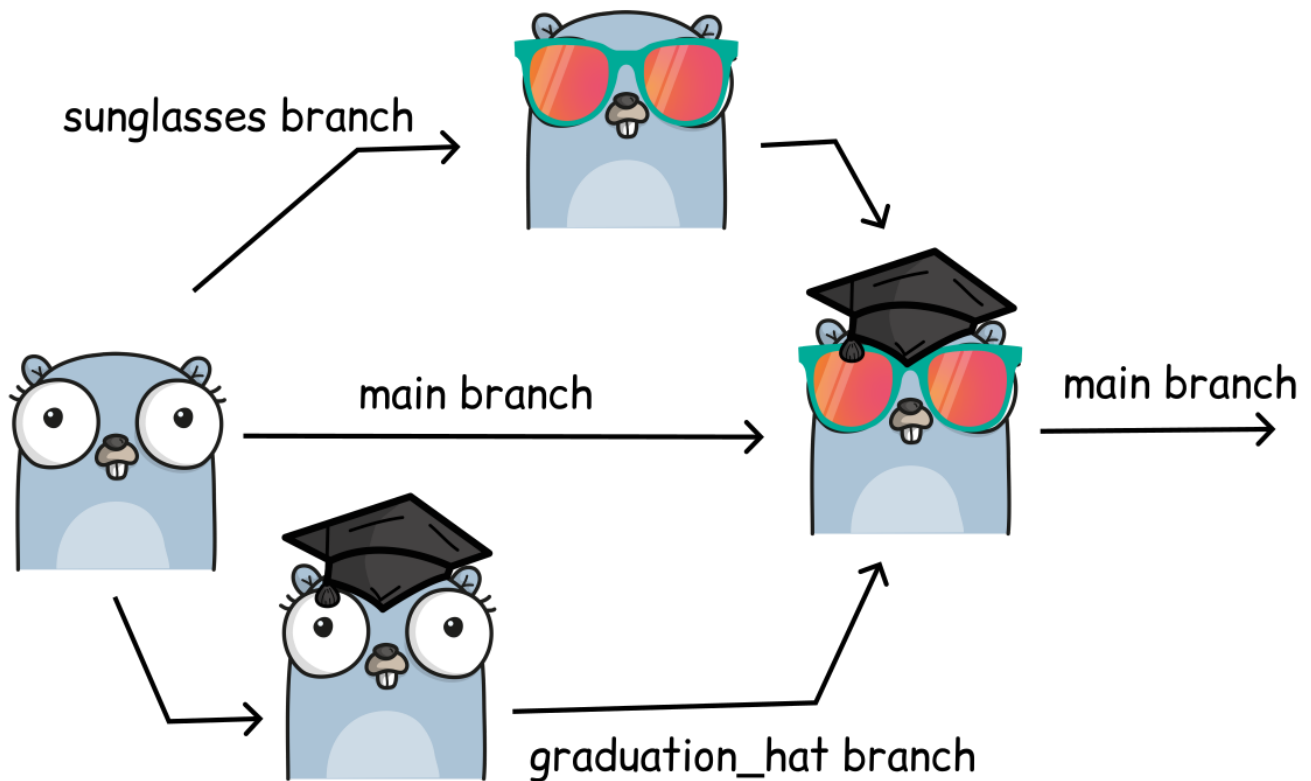
The first and most basic task to do in Git is **record changes** using commits. In this part, we will record changes in two ways: on a new branch (which supports multiple lines of work at once), and directly on the “main” branch (which happens to be the default branch here).

Objectives

- Record new changes to our own copy of the project.
- Understand adding changes in two separate branches.
- See how to compare different versions.

Background

- In the previous episode we have browsed an existing repository and saw commits and branches.
- Each commit is a snapshot of the entire project at a certain point in time and has a unique identifier (hash) .
- A branch is a line of development, and the `main` branch or `master` branch are often the default branch in Git.
- A branch in Git is like a sticky note that is attached to a commit. When we add new commits to a branch, the sticky note moves to the new commit.
- Tags are a way to mark a specific commit as important, for example a release version. They are also like a sticky note, but they don’t move when new commits are added.



What if two people, at the same time, make two different changes? Git can merge them together easily. Image created using <https://gopherize.me/> (inspiration).

Recording changes [2]

Exercise

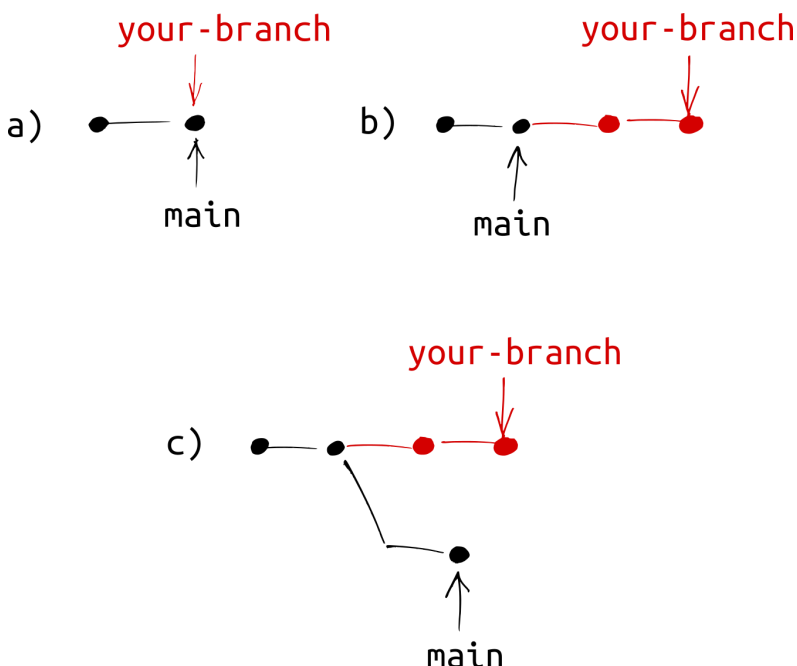


Illustration of what we want to achieve in this exercise.

We offer **four different paths** of how to do this exercise. For the CodeRefinery workshop day 1, we use and demonstrate the **GitHub path** only and recommend you do that (you can get experience with the other paths on day 2).



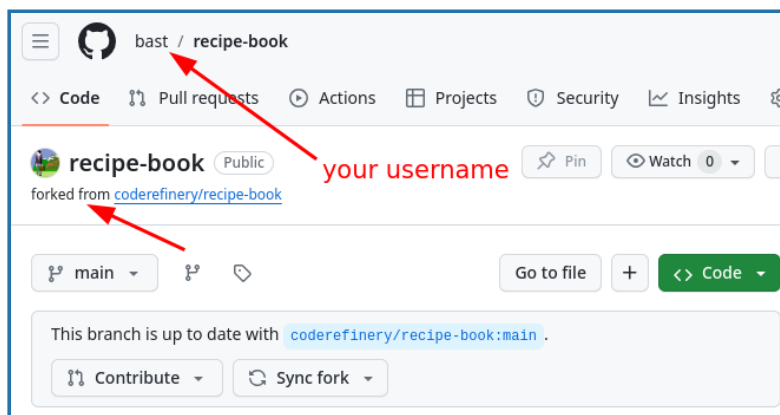
Exercise: Practice creating commits and branches (20 min)

1. Make sure that you now work **on your fork** of the recipe-book repository (`USER/recipe-book` , not `cr-workshop-exercises/recipe-book`)
2. First create a new branch and then add a recipe to the branch and commit the change.
3. In a new commit, modify the recipe you just added.
4. Switch to the `main` branch and modify a recipe there.
5. Browse the network and locate the commits that you just created (“Insights” -> “Network”).
6. Compare the branch that you created with the `main` branch. Can you find an easy way to see the differences?
7. Can you find a way to compare versions between two arbitrary commits in the repository?
8. Try to rename the branch that you created and then browse the network again.
9. Try to create a tag for one of the commits that you created (on GitHub, create a “release”).

The solution below goes over most of the answers, and you are encouraged to use it when the hints aren’t enough - this is by design.

Solution and walk-through

(1) Make sure you are on your fork



You want to see your username in the URL and you want to see the “forked from ...” part.

(2) Create a branch and add a recipe to the branch

A recipe template is below. This format is called “Markdown”, but it doesn’t matter right now. You don’t have to use this particular template.

Recipe name

Ingredients

- Ingredient 1
- Ingredient 2

Instructions

- Step 1
- Step 2

There is a main branch that is default. We want to create a *different* branch for our new commit, because we will *merge* it later. Commit is the verb to describe recording more changes, and also the name of the thing you make. A commit is identified by something such as `554c187`.

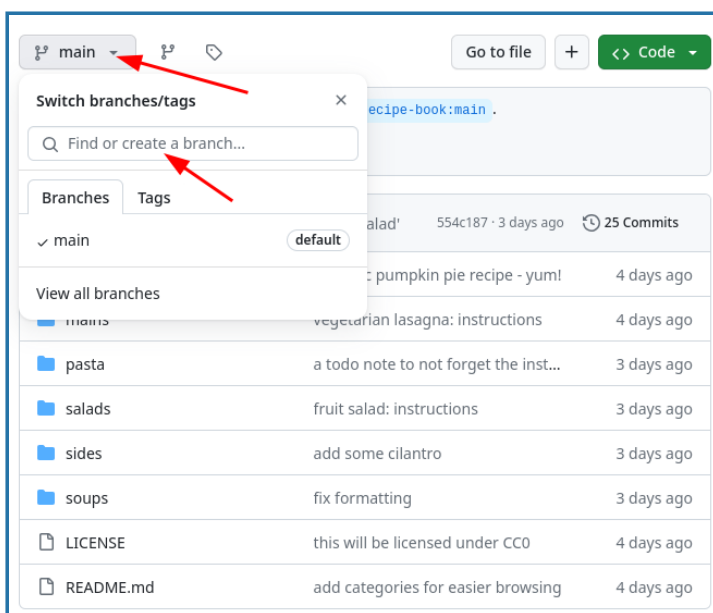
GitHub

VS Code

Command line

RStudio

1. Where it says “main” at the top left, click, enter a new branch name `new-recipe`, click on the offer to create the new branch (“Create branch new-recipe from main”).



2. Change to some sub-directory, for example `sides`.
3. Make sure you are still on the `new-recipe` branch (it should say it at the top), and click “Add file” → “Create new file” from the upper right.
4. Enter a filename where it says “Name your file...”, with a `.md` at the end. Example: `mixed-nuts.md`.
5. Enter the recipe. You can use the template above.
6. Click “Commit changes”
7. Enter a commit message. Then click “Commit changes”.

You should appear back at the file browser view, and see your new recipe there.

(3) Modify the recipe with a new commit

GitHub

VS Code

Command line

RStudio

This is similar to before, but we click on the existing file to modify.

1. Click on your new recipe, for example `mixed-nuts.md`.
2. Click the edit button, the pencil icon at top-right.
3. Follow the “Commit changes” instructions as in the previous step.

(4) Switch to the main branch and modify a recipe there

GitHub

VS Code

Command line

RStudio

1. Go back to the main repository page (your user’s page).
2. In the branch switch view (top left above the file view), switch to `main`.
3. Modify another recipe that already exists, following the pattern from above. Don’t modify the one you just created (but it shouldn’t even be visible on the `main` branch).

Modify another recipe and commit your changes.

(5) Browse the commits you just made

Let’s look at what we did. Now, the `main` and `new-recipe` branches have diverged: both have some modifications. Try to find the commits you created.

GitHub

VS Code

Command line

RStudio

Insights tab → Network view (just like we have done before).

(6) Compare the branches

Comparing changes is an important thing we need to do. When using the GitHub view only, this may not be so common, but we’ll show it so that it makes sense later on.

GitHub

VS Code

Command line

RStudio

A nice way to compare branches is to add `/compare` to the URL of the repository, for example (replace USER): `https://github.com/USER/recipe-book/compare`

(7) Compare two arbitrary commits

This is similar to above, but not only between branches.

GitHub

VS Code

Command line

RStudio

Following the `/compare`-trick above, one can compare commits on GitHub by adjusting the following URL: `https://github.com/USER/recipe-book/compare/VERSION1..VERSION2`

Replace `USER` with your username and `VERSION1` and `VERSION2` with a commit hash or branch name. Please try it out.

(8) Renaming a branch

GitHub

VS Code

Command line

RStudio

Branch button → View all branches → three dots at right side → Rename branch.

(9) Creating a tag

Tags are a way to mark a specific commit as important, for example a release version. They are also like a sticky note, but they don't move when new commits are added.

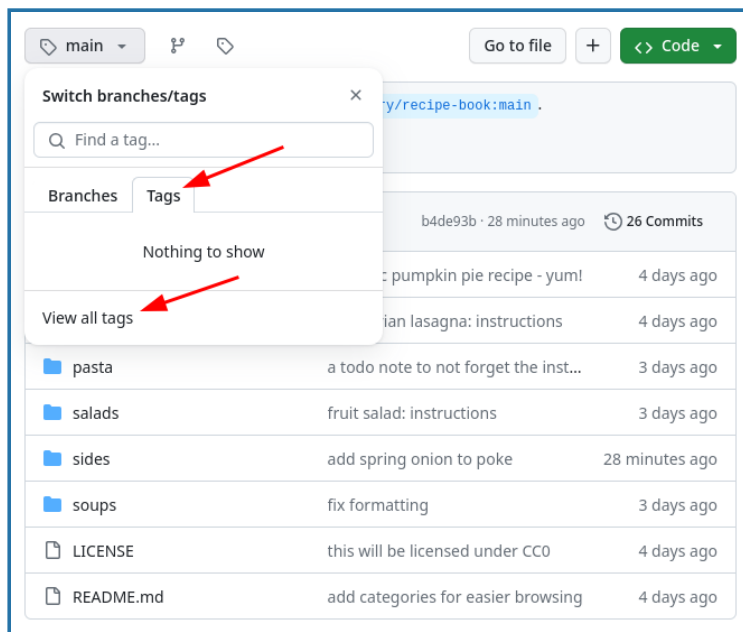
GitHub

VS Code

Command line

RStudio

Click on the branch switcher, and then on "Tags", then on "View all tags", then "Create a new release":



What GitHub calls releases are actually tags in Git with additional metadata. For the purpose of this exercise we can use them interchangeably.

Discussion

In this part, we saw how we can make changes to our files. With branches, we can track several lines of work at once, and can compare their differences.

- You could commit directly to `main` if there is only one single line of work and it's only you.
- You could commit to branches if there are multiple lines of work at once, and you don't want them to interfere with each other.
- Tags are useful to mark a specific commit as important, for example a release version.
- In Git, commits form a so-called "graph". Branches are tags in Git function like sticky notes that stick to specific commits. What this means for us is that it does not cost any significant disk space to create new branches.
- Not all files should be added to Git. For example, temporary files or files with sensitive information or files which are generated as part of the build process should not be added to Git. For this we use `.gitignore` (more about this later: what-to-avoid).
- Unsure on which branch you are or what state the repository is in? On the command line, use `git status` frequently to get a quick overview.

Merging changes and contributing to the project [1]

Git allows us to have different development lines where we can try things out. It also allows different people to work on the same project at the same. This means that we have to somehow combine the changes later. In this part we will practice this: merging.

Objectives

- Understand that on GitHub merging is done through a pull request. Think of it as a **change proposal**.
- Create and merge a pull request within your own repository.
- Understand (and optionally) do the same across repositories, to contribute to the upstream public repository.

Instructor note

- 10 min introduction and setup
- 25 min exercise
- 15 min discussion

Background

- In the last episode, we added a new recipe on a branch. This allows us to test it before it becomes “live”.
- Now, we want to bring that change into the “main” branch.
- We will find it’s not that hard! But you do have to keep track of the steps and make sure that you are careful about where a change is added.

Merging changes and contributing to the project [2]

Exercise

In this exercise, we will show how we can **propose changes** and **merge changes** within our own repository. Optionally, you can propose a recipe to the upstream recipe book - which shows the true purpose of this. But this is only a preview and we will practice collaboration much more in the [collaborative Git lesson](#).

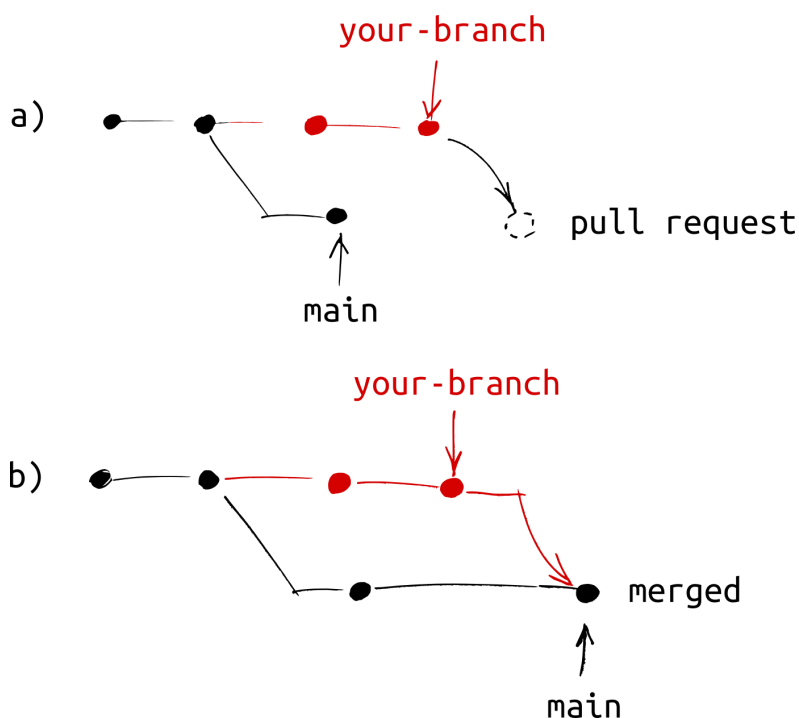


Illustration of what we want to achieve in this exercise.

We offer **four different paths** of how to do this exercise. For the CodeRefinery workshop day 1, we use and demonstrate the **GitHub path** only and recommend you do that. The exercise text below has some GitHub-specific notes, but most is possibly with any path.

GitHub

Local (VS Code, Command line, RStudio)

First, we make something called a pull request, which allows review and commenting before the actual merge.



Exercise: Merging branches with pull requests (20 min)

We assume that in the previous exercise you have created a new branch with a recipe. In our previous example, it is called `new-recipe`. If not, create the branch first and add a recipe to your new branch, see commits.

We provide basic hints. You should refer to the solution as needed.

1. Navigate to your branch from the previous episode (hint: the same branch view we used last time).
2. Begin the pull request process (hint: There is a “Contribute” button in the branch view).
3. Add or modify the pull request title and description, and verify the other data. In the pull request verify the target repository and the target branch. Make sure that you are merging within your own repository. **GitHub: By default, it will offer to make the change to the upstream repository, `cr-workshop-exercises`. You should change this,** you shouldn’t contribute your test recipe upstream yet. Where it says `base repository`, select your own user’s repository.
4. Create the pull request by clicking “Create pull request”. Browse the network view to see if anything has changed yet.
5. Merge the pull request, or if you are not on GitHub you can merge the branch locally. Browse the network again. What has changed?
6. Find out which branches are merged and thus safe to delete. Then remove them and verify that the commits are still there, only the branch labels are gone (hint: you can delete branches that have been merged into `main`).
7. Optional: Try to create a new branch with a new change, then open a pull request but towards the central repository. We will later merge few of those. (Hint: this is mostly the same as above, for the GitHub path. But, you set the base repository as CodeRefinery. You might need to compare across forks.)

The solution below goes over most of the answers, and you are encouraged to use it when the hints aren’t enough - this is by design.

Solution and walk-through

(1) Navigate to your branch

Before making the pull request, or doing a merge, it's important to make sure that you are on the right branch. Many people have been frustrated because they forgot this!

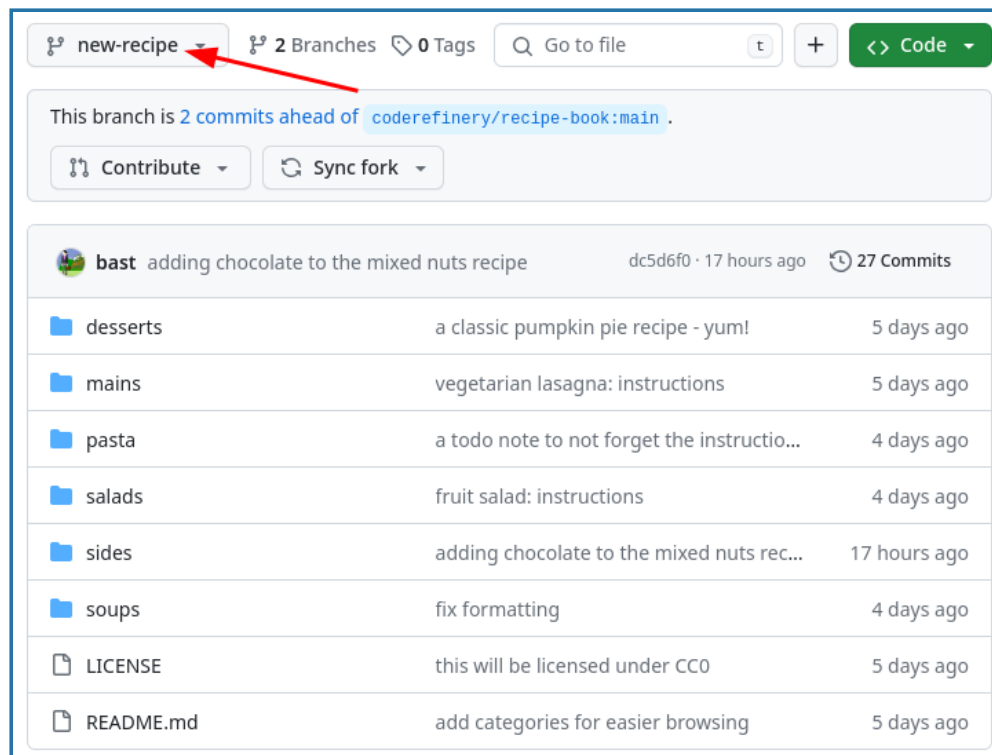
GitHub

VS Code

Command line

RStudio

On GitHub make sure you are on the branch you want to merge **from**:



The screenshot shows the GitHub interface for a repository named 'new-recipe'. At the top, the branch 'new-recipe' is selected, with a red arrow pointing to it. Below this, it states 'This branch is 2 commits ahead of coderefinery/recipe-book:main'. There are buttons for 'Contribute' and 'Sync fork'. The main content area shows a commit by 'bast' with the message 'adding chocolate to the mixed nuts recipe' and a list of files:

File	Commit Message	Time
desserts	a classic pumpkin pie recipe - yum!	5 days ago
mains	vegetarian lasagna: instructions	5 days ago
pasta	a todo note to not forget the instructio...	4 days ago
salads	fruit salad: instructions	4 days ago
sides	adding chocolate to the mixed nuts rec...	17 hours ago
soups	fix formatting	4 days ago
LICENSE	this will be licensed under CC0	5 days ago
README.md	add categories for easier browsing	5 days ago

(2) Begin the pull request process

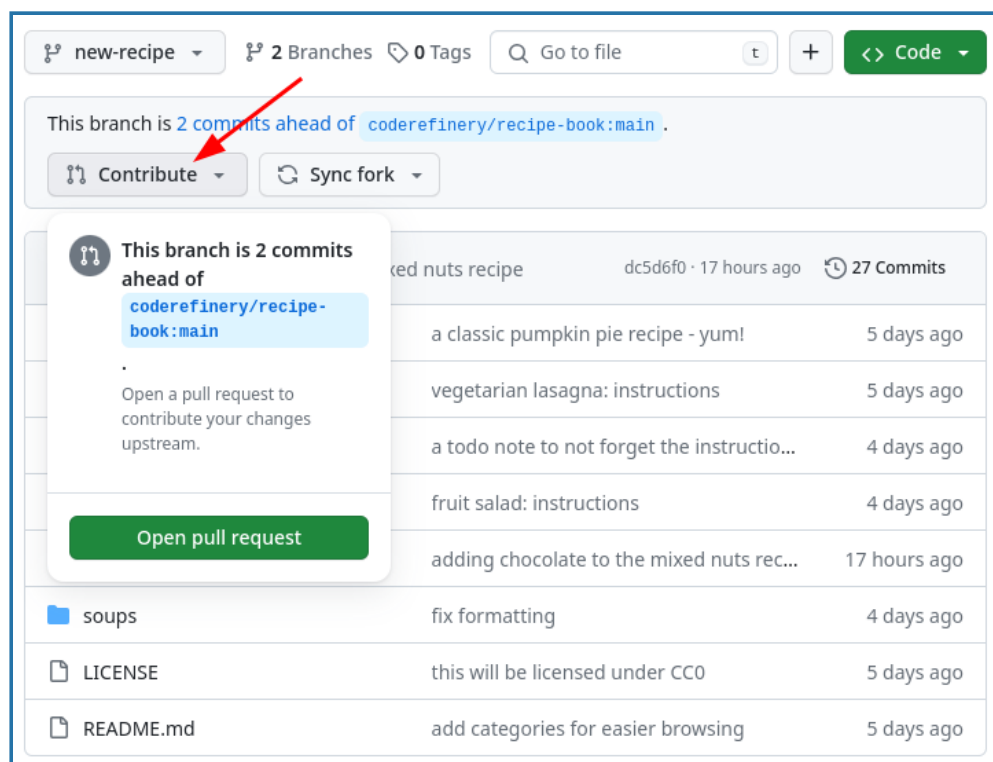
In GitHub, the pull request is the way we propose to merge two branches together. We start the process of making one.

GitHub

VS Code

Command line

RStudio



(3) Fill out and verify the pull request

Check that the pull request is directed to the right repository and branch and that it contains the changes that you meant to merge.

GitHub

VS Code

Command line

RStudio

Things to check:

- Base repository: this should be your own
- Title: make it descriptive
- Description: make it informative
- Scroll down to see commits: are these the ones you want to merge?
- Scroll down to see the changes: are these the ones you want to merge?

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare branches](#).

base: main ← **change this to your repository** compare: new-recipe ✓ **Able to merge.** These branches can be automatically merged.

Add a title

New recipe

Add a description

Write Preview H B I @

Add your description here...

Markdown is supported Paste, drop, or click to add files

Create pull request

This screenshot only shows the top part. If you scroll down, you can see the commits and the changes. We recommend to do this before clicking on “Create pull request”.

(4) Create the pull request

We actually create the pull request. Don't forget to navigate to the Network view after opening the pull request. Note that the changes proposed in the pull request are not yet merged.

GitHub

VS Code

Command line

RStudio

Click on the green button “Create pull request”.

If you click on the little arrow next to “Create pull request”, you can also see the option to “Create draft pull request”. This will be interesting later when collaborating with others. It allows you to open a pull request that is not ready to be merged yet, but you want to show it to others and get feedback.

(5) Merge the pull request

Now, we do the actual merging. We see some effects now.

Review it again (commits and changes), and then click “Merge pull request”.

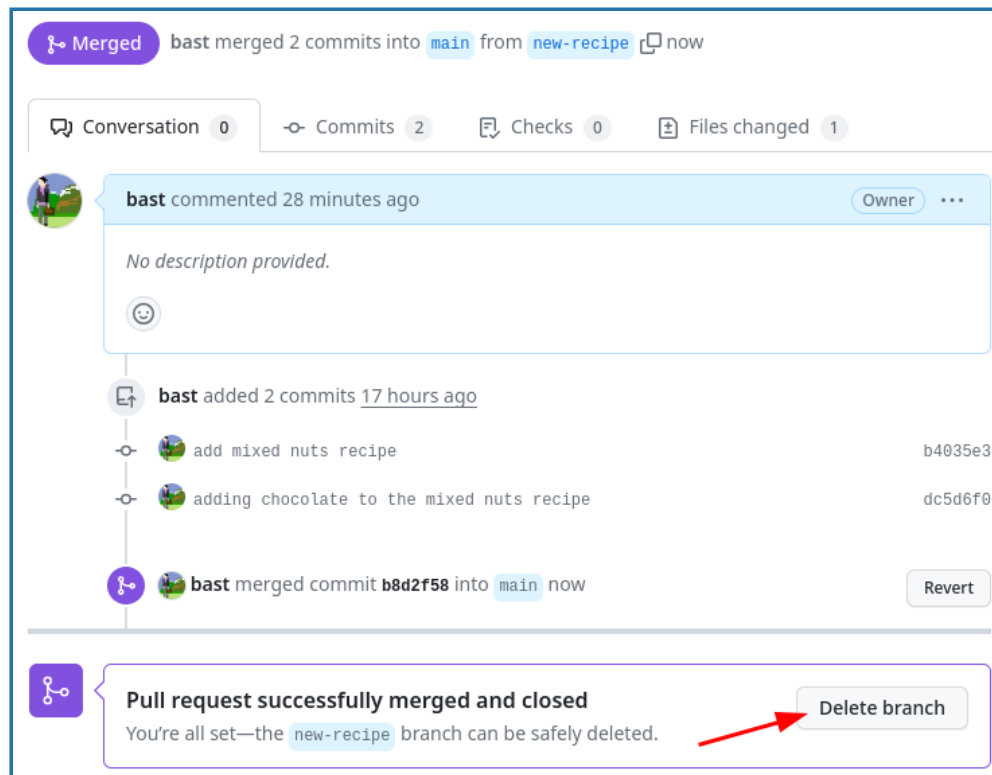
After merging, verify the network view. Also navigate then to your “main” branch and check that your new recipe is there.

(6) Delete merged branches

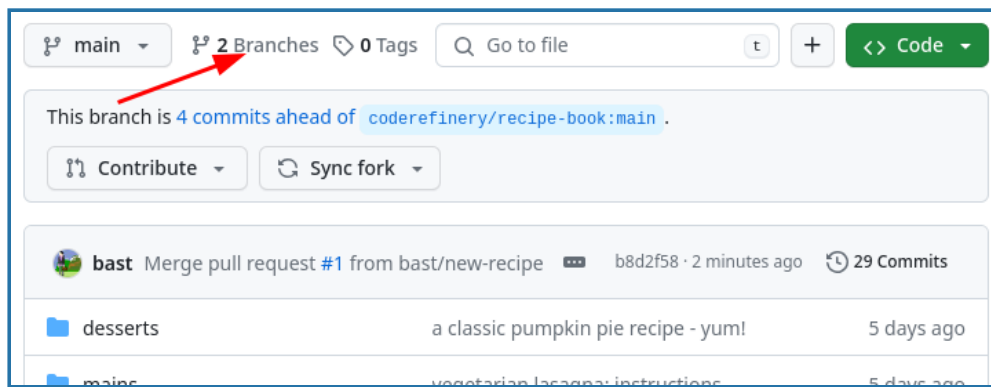
Before deleting branches, first check whether they are merged.

If you delete an un-merged branch, it will be difficult to find the commits that were on that branch. If you delete a merged branch, the commits are now also part of the branch where we have merged to.

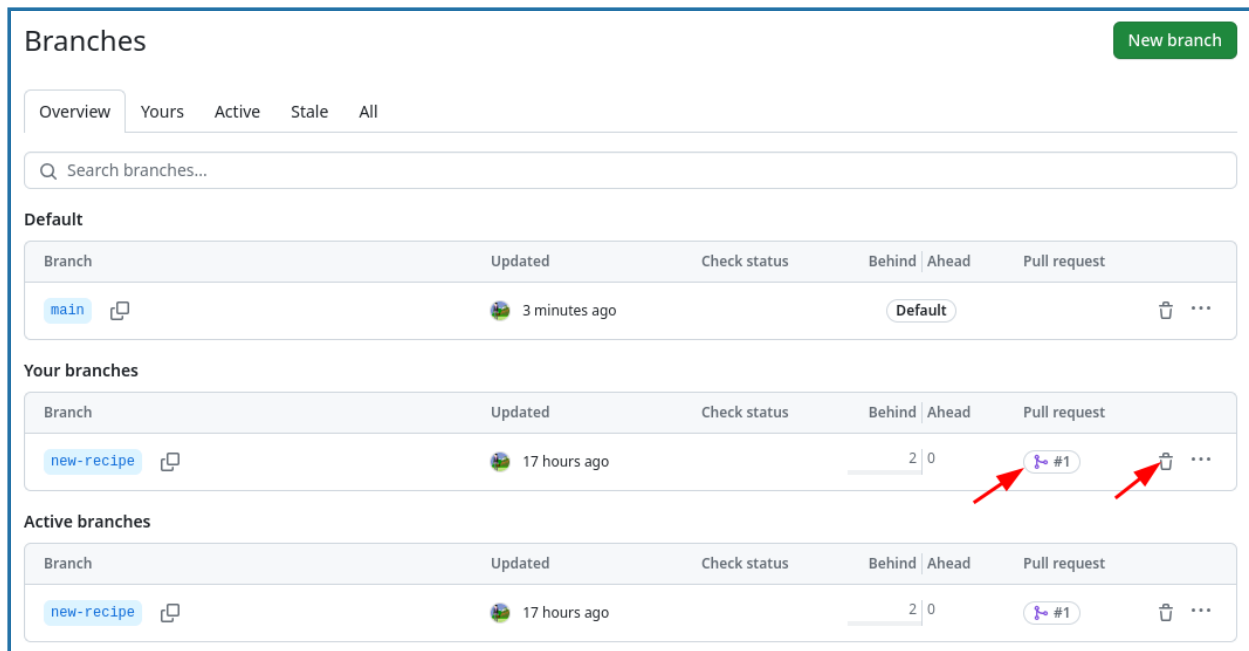
One way to delete the branch is to click on the “Delete branch” button after the pull request is merged:



But what if we forgot? Then navigate to the branch view:



In the overview we can see that it has been merged and we can delete it:



(7) Contribute to the original repository with a pull request

Remember, this is an advanced step. If you do this, you are donating a recipe to everyone.

GitHub

VS Code

Command line

RStudio

Now that you know how to create branches and opening a pull request, try to open a new pull request with a new change but this time the base repository should be the upstream one.

In other words, you now send a pull request across repositories: from your fork to the original repository.

Another thing that is different now is that you might not have permissions to merge the pull request. We can then together review and browse the pull request.

Resolving a conflict (demonstration)

A conflict is when Git asks humans to decide during a merge which of two changes to keep if the same portion of a file has been changed in two different ways on two different branches.

We will practice conflict resolution in the [collaborative Git lesson](#).

Here we will only demonstrate how to create a conflict and how to resolve it, all on GitHub. Once we understand how this works, we will be more confident to resolve conflicts also in the command line.

How to create a conflict (please try this in your own time *and just watch now*):

- Create a new branch from `main` and on it make a change to a file.
- On `main`, make a different change to the same part of the same file.
- Now try to merge the new branch to `main`. You will get a conflict.

How to resolve conflicts:

- On GitHub, you can resolve conflicts by clicking on the “Resolve conflicts” button. This will open a text editor where you can choose which changes to keep. Make sure to remove the conflict markers. After resolving the conflict, you can commit the changes and merge the pull request.
- Sometimes a conflict is between your change and somebody else’s change. In that case, you might have to discuss with the other person which changes to keep.

How to avoid conflicts:

- We will talk about it in the [collaborative Git lesson](#).

Summary

- We learned how to merge two branches together.
- When is this useful? This is not only useful to combine development lines in your own work. Being able to merge branches also forms a basis for collaboration.
- Branches which are merged to other branches are safe to delete, since we only delete the “sticky note” next to a commit, not the commits themselves.

Cloning a Git repository and working locally [1]

If you’ve been following the main path, you have just had interacted with repositories on GitHub. This might not be what you usually do, so now we move to working on your own computer.

- We are able to clone a repository from the web and modify it locally.
- We can do the same things we did before (commit, branch, merge), but **locally**.
- We get a feeling for remote repositories ([more later](#)).

Instructor note

- 10 min introduction and setup
- 25 min exercise
- 15 min discussion

What is in a Git repository and what are we cloning?

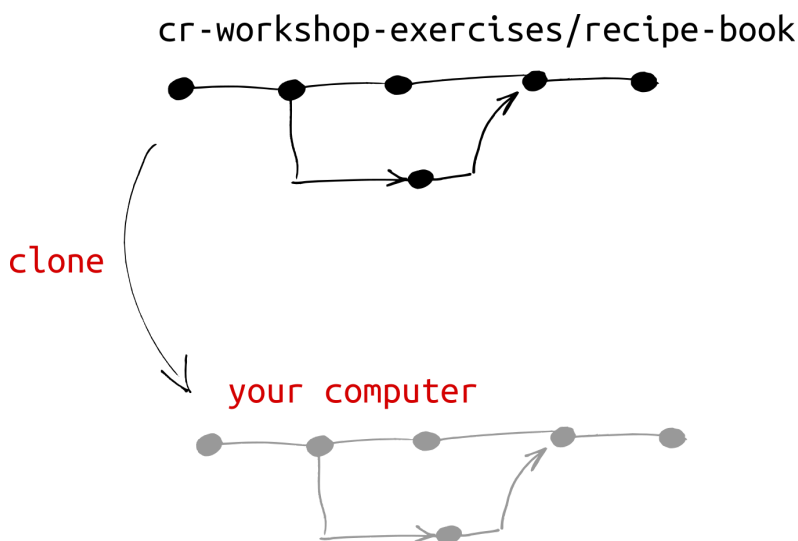


Illustration of cloning a repository to a local computer.

Git repository:

- Contains all the files and directories of a project.
- Contains the complete history of all changes (commits) to these files and directories.
- Each commit is a snapshot of the entire project at a certain point in time and has a unique identifier ("hash").
- Sometimes it contains multiple branches and tags.
- All the commits and history of a local repository are stored in a directory called `.git` which is located at the root of the repository.

Cloning:

- Copying (downloading) the entire repository with all commits, branches, and tags to your computer.
- It is a **full backup** of the repository, including all history.
- You can then work on your local clone of the repository.
- Changes on local clone will not automatically appear in the repository where we cloned from. We have to actively "push" them there (we will practice this in a later episode: sharing-repositories).

Cloning a Git repository and working locally [2]

Exercise

Work on this by yourself or in your teams. Conceptually this episode should seem familiar, from the browser-based exercises we did yesterday.

We offer the **Command Line**, **VS Code** and **RStudio** paths for this exercise. GitHub isn't an option in this episode, since that is what we already demonstrated in commits and merging and since the point of this episode is to work **locally**.

It is also possible to use the command line (terminal) from inside VS Code or RStudio.



Exercise: Cloning a Git repository and working locally (25 min)

1. [Configure Git command line and editor](#) if you haven't done that already.
2. **Decide which repository you want to clone:** your fork or the original repository? Both will work for this exercise. Then, **clone the recipe book**.
3. **Create a new branch.**
4. Make a **commit** on your new branch.
5. **Switch** back to the `main` branch and create one or two commits there.
6. **Merge** the new branch into `main`.
7. Compare the graph locally and on GitHub and observe that the **changes only exist locally on your computer**.
8. Where are the **remote branches**? Practice how you can see all remote branches also locally and how you can fetch them and make local changes to them.

The solution below goes over most of the answer and should be used as your guide (you can't figure it out just from the exercise instructions).

Solution and walk-through

(1) Configure Git command line and editor

We have an own section for this: [Configuring Git command line and editor](#).

(2) Cloning a repository

Now you need to decide which repository you want to clone. All of these options will work for this exercise since we don't plan to push changes back (for step 8 it might be easier to use the original repository):

- Clone the recipe book from your fork.

- Or clone the recipe book from the original repository: <https://github.com/cr-workshop-exercises/recipe-book>
- Or first fork the original repository and then clone your fork.

The examples below assume you are cloning the original repository. If you are cloning your fork, you should replace `cr-workshop-exercises` with your GitHub username.

Command line

VS Code

RStudio

If you are unsure whether you are using SSH or HTTPS, please read [Authenticating to GitHub: SSH or HTTPS or VS Code or RStudio?](#).

SSH

HTTPS

```
$ git clone git@github.com:cr-workshop-exercises/recipe-book.git
```

This creates a directory called “recipe-book” unless it already exists. You can also specify the target directory on your computer (in this case “my-recipe-book”):

SSH

HTTPS

```
$ git clone git@github.com:cr-workshop-exercises/recipe-book.git my-recipe-book
```

(3) Creating branches locally

Command line

VS Code

RStudio

Create a new branch called `another-recipe` from `main` and switch to it:

```
$ git switch --create another-recipe main
```

If you leave out the last argument, it will create a branch from the current branch:

```
$ git switch --create another-recipe
```

(4) Creating commits locally

Command line

VS Code

RStudio

Create a new file. **After we have created it**, we can stage and commit the change:

```
$ git add new-file.md  
$ git commit -m "Short summary of the change"
```

Make sure to replace “new-file.md” with the actual name of the file you created and to replace “Short summary of the change” with a meaningful commit message.

(5) Switching branches and creating commits

Command line

VS Code

RStudio

First switch to the `main` branch:

```
$ git switch main
```

Then modify a file. Finally `git add` and then commit the change:

```
$ git commit -m "Short summary of the change"
```

Modify another recipe and commit your changes.

(6) Merging branches locally

Command line

VS Code

RStudio

On the command line, when we merge, we always modify our current branch.

If you are not sure anymore what your current branch is, type:

```
$ git branch
```

Another way to find out where we are in Git:

```
$ git status
```

In this case we merge the `another-recipe` branch into our current branch:

```
$ git merge another-recipe
```

(7) How to compare the graph locally and on GitHub

Command line

VS Code

RStudio

```
$ git log --graph --oneline --decorate --all
```

We recommend to define an alias in Git, to be able to nicely visualize branch structure in the terminal without having to remember a long Git command:

```
$ git config --global alias.graph "log --all --graph --decorate --oneline"
```

Then you can just type `git graph` from there on. We have an own section about aliases: aliases.

Compare this with the graph on GitHub: Insights tab → Network view (just like we have done before). The result is that we should not be able to see the new branch and the new commits on GitHub (since we haven't pushed it to GitHub yet - it is only local work so far).

(8) Browsing remote branches and creating local branches from them

With `git branch` you can list all local branches:

```
$ git branch  
  
  another-recipe  
* main
```

But where are the remote branches? We expect to see [a couple of them](#).

We can see them by asking for all branches (your output might vary depending on where you cloned from):

```
$ git branch --all  
  
  another-recipe  
* main  
remotes/origin/HEAD -> origin/main  
remotes/origin/alex/fruit-salad  
remotes/origin/main  
remotes/origin/radovan/lasagna  
remotes/origin/radovan/poke
```

You can create a local branch from a remote branch which will “track” the remote branch. For instance, to create a local branch `alex/fruit-salad` from the remote branch `origin/alex/fruit-salad` and switch to it, you can do:

```
$ git switch --create alex/fruit-salad origin/alex/fruit-salad
```

This shortcut will do the same thing:

```
$ git switch --track origin/alex/fruit-salad
```

Or even shorter:

```
$ git switch alex/fruit-salad
```

If you want to create a branch and not switch to it, you can use `git branch`.

```
$ git branch alex/fruit-salad
```

Summary

- When we clone a repository, we get a full backup of the repository, including all history: all commits, branches, and tags.
- Yesterday we learned about branches and commits, and now we created and used them locally.
- **Creating local branches and commits does not automatically modify the remote repository.** To “push” our local changes to the remote repository, we have to actively “push” them there. We will practice this in a later episode: sharing-repositories
- Remote branches and local branches are not the same thing. If we want to create local commits, we always need to create a local branch first. But the local branch can “track” the remote branch and we can push and pull changes to and from the remote branch.